THE NOVA RESEARCH CURRICULUM

# Mathematics

Grade 7 · Term 3

Student edition

Outcomes: MA-7.6 circumference and area of circles, surface area and volume of prisms and cylinders; MA-7.7 histograms, line graphs, grouped mean, median, mode and probability.

## Lesson notes

- **Circles (Weeks 1–4):** circumference = $2\pi r$ (or $\pi d$); area = $\pi r^2$; use $\pi \approx 22/7$ or 3.14. Measure round objects with string to discover π.

  ▶ **Try it on NovaSchool:** Unrolling a circle: where π comes from — open **Simulations** on your dashboard.

- **Prisms and cylinders (Weeks 5–8):** volume of a prism = area of cross-section × length; volume of a cylinder = $\pi r^2 h$; surface area of a cylinder = $2\pi r^2 + 2\pi rh$. Nets of prisms and cylinders.

  ▶ **Try it on NovaSchool:** Folding a net into a solid — open **Simulations** on your dashboard.

- **MA-7.7 Data (Weeks 9–12):** grouped frequency tables; histograms (bars touch); line graphs. Grouped mean = Σ(midpoint × frequency) ÷ total frequency; modal class; median class. Probability of an event = favourable outcomes ÷ total outcomes; the probabilities of all outcomes add to 1.

## Workbook pages

| Page | Task |
|---|---|
| 1–2 | Circumference and area of 10 circles. |
| 3–4 | Volume and surface area of prisms and cylinders; draw two nets. |
| 5–6 | Grouped data from the class survey: histogram, mean, modal class; bag-of-counters probability. |

## Question bank

| # | Outcome | Question | Your answer |
|---|---|---|---|
| 1 | MA-7.6 | Circumference of a circle with radius 7 cm (π = 22/7)? | 22 cm / 44 cm / 154 cm / 49 cm |
| 2 | MA-7.6 | Area of a circle with radius 7 cm (π = 22/7)? | 44 cm² / 49 cm² / 154 cm² / 308 cm² |
| 3 | MA-7.6 | Volume of a cylinder with r = 7 cm, h = 10 cm (π = 22/7)? | 440 cm³ / 1,540 cm³ / 154 cm³ / 4,400 cm³ |
| 4 | MA-7.6 | A prism with cross-section area 12 cm² and length 5 cm has volume … | 17 cm³ / 60 cm³ / 120 cm³ / 60 cm² |
| 5 | MA-7.6 | Doubling the radius of a circle doubles its area. True or false. | True / False |

| # | Outcome | Question | Your answer |
|---|---|---|---|
| 6 | MA-7.7 | Classes 0–10 (f = 2), 10–20 (f = 3), 20–30 (f = 5). Grouped mean? | 15 / 18 / 20 / 25 |
| 7 | MA-7.7 | In the same table, the modal class is … | 0–10 / 10–20 / 20–30 / none |
| 8 | MA-7.7 | In a histogram the bars … | have gaps / touch / are circles / are always the same height |
| 9 | MA-7.7 | A bag has 3 red and 5 blue counters. P(red)? | 3/5 / 3/8 / 5/8 / 1/3 |
| 10 | MA-7.7 | If P(rain) = 0.3, then P(no rain) = ? | 0.3 / 0.7 / 1.3 / 0 |